

Alexandre Benoit

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Summary

ML researcher and engineer with 2+ years of research experience specialising in building, training, and debugging complex ML systems at scale. 8 peer-reviewed publications across the University of Cambridge, UCL, and Bath spanning low-precision deep learning on accelerators, graph neural networks, semi-supervised learning, multi-agent reinforcement learning, and model interpretability. Hands-on experience with GPU cluster training, mixed-precision training (BF16/FP16/FP32), large-scale synthetic data pipelines, and systematic model benchmarking with Weights & Biases. Open-source contributor for Stripe and Tempo on the Machine Payments Protocol for AI agents (npm). Strong software engineering foundation in Python, PyTorch, C/C++, and TypeScript, with production deployment experience using Docker, CI/CD, and Azure.

Experience

Savanta, ML/AI Engineer & Solution Architect

Architect and build Virtual Personas, Synthetic Personas and ACMD (Automatic Clustering of Market Data), production AI agentic orchestration and unsupervised machine learning platforms for market research. Built an unsupervised clustering market-segmentation pipeline replacing 1-3 months of work in hours, a synthetic-respondent generation engine, and a society-simulation system, all driven by LLM agents. Own the full stack across six microservices (FastAPI, React, TypeScript, PostgreSQL, MongoDB, Redis) on Azure, with agent pipelines on LangChain, LangGraph, and Langfuse routing across OpenAI and Anthropic models. Built evaluation harnesses validating outputs against ground-truth distributions, ran 100+ prompt-fidelity benchmarks, and led a full security audit. **The products I built have generated just under £1M in client revenue in their first 5 months.** Build interactive dashboards and present findings directly to non-technical and C-suite stakeholders at enterprise clients (JP Morgan, Chase, Brompton, BPP, LVMH); coordinate delivery across a small cross-functional team.

London, UK

Jan 2026 – Present

University of Cambridge, ML Engineer Research Assistant — Lab of Prof. José Miguel Hernández-Lobato
Developed low-precision execution policies for the MACE equivariant GNN (molecular dynamics force fields). Profiled per-block, compared e3nn vs NVIDIA cuEquivariance backends, and achieved $\sim 12\times$ combined inference speedup (cuEquivariance + BF16/FP16 linear layers) while preserving physical fidelity. Published: [Speeding Up MACE: Low-Precision Tricks for Equivariant Force Fields](#).

Cambridge, UK

May 2025 – Sept 2025

University College London (UCL), ML Engineer Research Assistant — e-Motion Lab, Dr. Pedram Asef
Designed a state-of-the-art semi-supervised deep kernel learning framework for battery state-of-health estimation (Encoder-T + GRU + GPR with custom kernel), beating the prior NASA benchmark by 24% MAPE. Accepted at *Energy and AI*. Trained on GPU clusters with Bayesian HPO via Weights & Biases.

London, UK

Jun 2023 – Present

University of Bath, Machine Learning Engineer — Research Assistant

Advanced-analytics research on energy & climate and autonomous systems (3 publications): Google OR-Tools and ML for global path planning (operations research); multi-agent RL for optimising commercial EV charging-station scheduling; and AI-driven flicker estimation in power systems. Built end-to-end pipelines for time-series forecasting, optimisation, and RL environment design. Built end-to-end ML pipelines for time-series prediction, RL environment design, and model evaluation.

Bath, UK

Jun 2023 – Sept 2024

Team Bath Racing Electric, Software Engineer — Autonomous Driving

Built autonomous-driving software in C++ and Python: MPC algorithms, MCP integration of AI computer-vision tools for the perception pipeline, and API integration with the Dynamic Driving Task system.

Bath, UK

Sept 2022 – Jun 2023

Jaguar TCS Formula E, Software Engineer

Built large-scale real-time driving simulation environments in C++ and MATLAB for race strategy R&D, with telemetry-driven feedback and a custom web interface for engineers.

Oxfordshire, UK

Sept 2021 – May 2022

Education

M.Phil, University of Cambridge, Clare College — Machine Learning and Machine Intelligence (MLMI). GPA: A-.

Cambridge, UK

Sept 2024 – Sept 2025

M.Eng, University of Bath — Integrated Mechanical & Electrical Engineering. First Class Honours (73%), top 2 of cohort, GPA 4.0/4.0.

Bath, UK

Sept 2019 – June 2024

Projects & Publications

Renewable-Energy Asset Originator (in progress) — Built an agentic data pipeline over public UK renewable-energy asset data: agents execute a deterministic screening plan against an input investment thesis to surface and validate origination leads.

8 peer-reviewed publications across Cambridge, UCL, and Bath in *Energy and AI*, *Advanced Intelligent Systems*, and the *Learning on Graphs Conference*. Full list: [Google Scholar](#).

Technical Skills

Languages: Python • C/C++ • TypeScript • JavaScript • MATLAB • SQL **ML:** PyTorch • TensorFlow • scikit-learn • JAX (familiar)

Gen AI & Agents: LangChain • LangGraph • Langfuse • OpenAI/Anthropic APIs • MCP • RAG • LLM Benchmarking • Multi-agent Workflows

Engineering: FastAPI • React • Docker • CI/CD (GitHub Actions) • Azure • PostgreSQL • MongoDB • Redis • Git • Railway • optimisation • statistics • data visualisation & dashboards **Languages:** French (Native), English (Fluent), Spanish (C1)